

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int numberOfProcesses;
6      printf("Enter number of processes: ");
7      scanf("%d", &numberOfProcesses);
8
9      int timeQuantum;
10     printf("Enter time quantum: ");
11     scanf("%d", &timeQuantum);
12
13     int burstTime[10];
14     int remainingTime[10];
15     int waitingTime[10] = {0};
16     int turnaroundTime[10];
17
18     // Read burst times
19     for(int i = 0; i < numberOfProcesses; i++)
20     {
21         printf("Enter burst time of process P%d: ", i + 1);
22         scanf("%d", &burstTime[i]);
23         remainingTime[i] = burstTime[i];
24     }
25
26     int currentTime = 0;
27     int allDone;
28
29     // Round Robin execution
30     do
31     {
32         allDone = 1;
33
34         for(int i = 0; i < numberOfProcesses; i++)
35         {
36             if(remainingTime[i] > 0)
37             {
38                 allDone = 0;
39
40                 if(remainingTime[i] > timeQuantum)
41                 {
42                     currentTime += timeQuantum;
43                     remainingTime[i] -= timeQuantum;
44                 }
45                 else
46                 {
47                     currentTime += remainingTime[i];
48                     waitingTime[i] = currentTime - burstTime[i];
49                     remainingTime[i] = 0;
50                 }
51             }
52         }
53     } while(!allDone);
54
55     // Calculate turnaround time
56     for(int i = 0; i < numberOfProcesses; i++)
57     {
58         turnaroundTime[i] = waitingTime[i] + burstTime[i];
59     }
60
61     // Output
62     printf("\nPID\tBT\tWT\tTAT\n");
63     for(int i = 0; i < numberOfProcesses; i++)
64     {
65         printf("P%d\t%d\t%d\t%d\n",
66             i + 1,
67             burstTime[i],
68             waitingTime[i],
69             turnaroundTime[i]);
70     }
71
72     return 0;
73 }
74
```